

Do Verifiers Reduce Security-Relevant Misconfigurations? A Preregistered Paired Evaluation of Verifier-in-the-Loop GVR for Intent-to-Configuration

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Abstract—We preregistered and executed a deterministic paired experiment to test whether embedding a reachability/policy verifier inside a generate–verify–repair (GVR) loop reduces security-increasing (forbidden-reachability) misconfigurations produced by an LLM-driven intent→configuration pipeline. Using Dockerized FRRouting-style emulation, two deterministic validators combined with a logical-OR scoring rule, and a matched sample of $N = 300$ intents (150 benign + 150 programmatic adversarial variants), we compared single-pass LLM generation (baseline) to an LLM+verifier GVR pipeline implemented under a frozen protocol (preregistration cnd-2026-netops-gvr-security-01-prereg-v1). The adapter used shared_initial_candidate semantics: one initial LLM candidate was generated per paired intent and the same candidate was replayed to both arms (baseline is therefore not an independent-generation condition). The run completed all planned episodes with model_calls_used = 300 (shared candidate). Confirmatory outcome: baseline_success = 300 and guarded_success = 300 (contingency $n_{11} = 300$, $n_{10} = n_{01} = n_{00} = 0$); validators never reported forbidden reachability and the repair step was never invoked. Paired risk-difference = 0.0, McNemar two-sided $p = 1.0$, 95% bootstrap CI = [0.0, 0.0] (10,000 resamples, seed 1729). Observing 0/300 violations gives a one-sided Clopper–Pearson 95% upper bound ≈ 0.00994 on an undetected baseline violation rate. Because the repair path was not exercised, this study quantifies that inserting a verifier does not change observed compliance when baseline generation already satisfies the safety predicate under the chosen contract; it does not provide evidence that verifier-guided repair can rescue genuine violations. We archive the preregistration, deterministic execution accounting, per-episode validator traces, and analysis artifacts to support reproducibility and to guide follow-up stress tests designed to exercise the repair path.

I. INTRODUCTION

Large language models (LLMs) provide practical routes to automate intent-driven network configuration, but operators reasonably fear that generated artifacts can expand attack surface, violate access-control policies, or otherwise introduce security-relevant regressions. Systems research and surveys therefore emphasize constrained generation, execution oracles, and iterative generate–verify–repair (GVR) loops as operational mitigations (see the cited literature). However, rigorous, preregistered confirmatory evidence that a verifier placed in a GVR loop meaningfully reduces forbidden-reachability misconfigurations under adversarial inputs is scarce.

We designed, preregistered, and executed a paired experiment to answer a focused operational question: Does embedding a reachability/policy verifier inside a GVR loop reduce the incidence of forbidden-reachability misconfigurations produced by an LLM-driven intent→configuration pipeline? Our study makes four evidence-grounded contributions: (1) a machine-readable preregistration that fixed a paired $N = 300$ confirmatory design; (2) a deterministic Dockerized execution that used two deterministic validators combined with a logical-OR decision rule and a shared_initial_candidate generation contract to conserve model calls; (3) full execution and analysis artifacts showing a degenerate confirmatory result ($n_{11} = 300$) together with a descriptive Clopper–Pearson upper bound on undetected baseline violation rate (≈ 0.00994); and (4) artifact-grounded operational recommendations and reproducibility materials to inform follow-up stress tests that intentionally exercise the repair path. All experimental artifacts (preregistration, execution manifest, per-episode validator traces, raw responses, and analysis outputs) are archived in the evidence bundle to enable independent verification.

II. RELATED WORK

Three literatures shaped our design and interpretation. First, prior GVR and execution-guided repair architectures have improved correctness on code and Infrastructure-as-Code tasks by using verifier feedback to direct repairs; such systems show that verifier power and repair strategy determine realized gains. Second, surveys and NetOps system papers document LLM failure modes—hallucination, overconfidence, and domain mismatch—and consistently recommend integrating validators, constrained generation, or human oversight to reduce operational risk. Third, emulation and testbed work demonstrates that execution-based validation is feasible for research-grade evaluation but warns of external-validity gaps when porting to production devices or live traffic.

Existing work rarely combines a preregistered paired evaluation, deterministic validators, and archived per-episode verifier traces aimed specifically at security-relevant reachability predicates. Our study fills that niche by fixing generation semantics, explicitly trading provider-call budget for deterministic

accounting, and archiving the deterministic artifacts needed for audit and targeted follow-up stress testing.

III. METHODOLOGY

Design and estimand: The preregistration specified a paired binary design with $N = 300$ matched intents (150 benign seeds plus 150 programmatic adversarial variants). The primary estimand is the `paired_success_rate_difference_guarded_minus_baseline` (absolute difference in the proportion of intents yielding forbidden reachability). The prespecified test was an exact McNemar two-sided test ($\alpha = 0.05$); confidence intervals were to be obtained by a bootstrap percentile method. Sensitivity checks treating technical failures as violations were prespecified; no technical failures occurred.

Generation contract and adapter semantics: To limit model-call cost under the execution contract we used `shared_initial_candidate` semantics: for each paired intent a single LLM candidate was generated and the same candidate was replayed to both arms (baseline and guarded). This yields exactly one hosted-model generation per pair (`model_calls_used = 300` for 300 pairs) rather than two independent draws per pair (which would consume up to 600 hosted-model calls). Because the baseline condition is a replay of the shared initial candidate rather than an independent draw, baseline is not an independent-generation condition; this reduces between-arm sample variance and the probability of observing discordant violating outputs across the pair, a design feature that directly affects paired-test sensitivity for repair efficacy.

Intervention arms and repair policy: Baseline = replay of the shared initial candidate. Guarded = the same shared initial candidate validated by two deterministic validators; if any validator reported forbidden reachability the preregistered repair policy allowed up to one repair LLM call (maximum one repair per pair). Repair attempts were only triggered by a validator-detected forbidden-reachability predicate.

Validators, scoring rule, and determinism: We evaluated two deterministic validators (a containerized execution-style reachability validator and a lightweight static reachability analyzer) and combined their outputs with a logical-OR decision rule: a pair was labeled violating if any validator reported the forbidden-reachability predicate. The emulation environment and all sampler and orchestration seeds were fixed to ensure deterministic replay of the experiment under the same contract. All per-episode validator judgments, final/initial configuration parses, and `repair_applied` indicators were archived.

IV. RESULTS

Execution and accounting: The run completed the planned episodes = 600 (300 pairs $\times$ 2 conditions) with completed = 600 and failed episodes = 0. Under the `shared_initial_candidate` contract the provider-call accounting shows `model_calls_used = 300`. The run interval and summary accounting are archived in the execution manifest within the evidence bundle.

Primary confirmatory outcome: All 300 complete pairs satisfied the forbidden-reachability predicate in both arms: `baseline_success_count = 300` and `guarded_success_count = 300`. The paired contingency table is $n_{11} = 300$, $n_{10} = 0$, $n_{01} = 0$, $n_{00} = 0$ (Table 1). The paired risk-difference (guarded - baseline) = 0.0; McNemar exact two-sided $p = 1.0$. The prespecified bootstrap-percentile 95% CI (10,000 resamples, seed 1729) is [0.0, 0.0].

Degenerate contingency and descriptive bound: The absence of discordant pairs ($n_{10} = n_{01} = 0$) makes the McNemar test uninformative about repair efficacy because no disagreements exist for the paired binary comparison. This degenerate outcome is a direct consequence of two coupled design facts: (a) `shared_initial_candidate` semantics reduced independent-sample variance between arms, and (b) none of the programmatic adversarial variants produced a validator-detectable forbidden reachability for the chosen model and validators. As a descriptive constraint, observing 0 violations in $N = 300$ independent episodes yields a one-sided Clopper-Pearson 95% upper bound on the baseline violation probability of $1 - 0.05^{1/300} \approx 0.00994$ ($\approx 0.994\%$).

Repair activity and validator concordance: Because validators never reported forbidden reachability on any initial candidate, no repair call was invoked and all per-episode `repair_applied` flags are false. Validator judgments were concordant across the two deterministic validators for every archived episode. Representative per-episode traces illustrating typical tasks and validator outputs are archived in the evidence bundle.

V. DISCUSSION

What the evidence supports: Under the exact adapter contract (`shared_initial_candidate`), declared model (gpt-5-mini), the evaluated deterministic validator set combined by logical OR, the emulator, and the programmatic adversarial perturbations used here, baseline generation satisfied the forbidden-reachability predicate for every sampled intent; therefore inserting the verifier did not change observed compliance. The archived artifacts consistently document provider-call accounting, per-episode validator traces, and the paired contingency supporting that statement.

What the evidence does not show: Because the repair path was never invoked, this experiment provides no evidence that verifier-guided repair can rescue violating initial candidates. The preregistered power calculation assumed a nonzero baseline violation rate and targeted detecting a 30 percentage-point absolute reduction; the degenerate observed contingency violates that assumption and renders the confirmatory McNemar inference moot with respect to repair efficacy. The null paired difference therefore should not be interpreted as evidence that verifiers are unnecessary in other deployments.

Design lessons and operational recommendations grounded in the run: (1) **Generation semantics matter:** `shared_initial_candidate` reduces hosted-model cost but also reduces independent-sample variability that could produce discordant failures the repair step might correct. For

experiments intended to exercise repair efficacy, independent-condition generation (two LLM draws per pair) or larger repair-call budgets are preferable. (2) Adversary potency matters: the deterministic adversarial variants used here did not trigger validator-detectable violations for the chosen model and validators; stronger or targeted adversaries against the specific verifier predicates should be used in follow-up stress tests. (3) Verifier strength matters: we evaluated two deterministic validators combined by logical OR; broader oracle diversity and production-grade verifiers can expose different failure modes. (4) Reproducible accounting is practical: deterministic execution, fixed seeds, and archived per-episode validator traces permit exact auditing of the observed contingency and provider-call budget trade-offs.

Threats to validity (artifact-grounded): Internal validity is limited by the `shared_initial_candidate` semantics and single-model scope. Construct validity depends on whether the forbidden-reachability predicate and the two deterministic validators adequately capture operational security risk. External validity is limited because tasks were synthetic and executed in deterministic emulation; results may not generalize to heterogeneous production devices, live traffic, or different operational constraints. Conclusion validity is constrained because the observed contingency is degenerate and the preregistered power assumptions are unmet. These limitations motivate targeted follow-up tests rather than reanalysis of the present confirmatory outcome.

VI. LIMITATIONS

- Degenerate confirmatory outcome: zero discordant pairs ($n_{11} = 300$, $n_{10} = n_{01} = n_{00} = 0$) render McNemar inference uninformative about repair efficacy when baseline violation rates are nonzero; the preregistered power calculation assumed a nonzero baseline violation rate and targeted detecting a 30 percentage-point absolute reduction.
- GVR not exercised: under the chosen task set, emulator, model, and validators the verifier never reported forbidden reachability and consequently the guarded repair step was never invoked; the experiment therefore tests whether verifier insertion changes compliance when baseline is already compliant, not whether verifier-guided repair rescues violations.
- Generation contract choice: the preregistered `shared_initial_candidate` semantics produced one LLM call per pair and reduced model-call overhead but also limited independent draws that could produce discordant errors; alternative contracts (independent-condition generation or expanded repair budgets) would better exercise repair paths.
- Adversarial-perturbation detail: the preregistration specified programmatic generation of one adversarial variant per benign seed; per-task generation seeds and intent payloads are archived in the evidence bundle for reproducibility.

- External validity: tasks were synthetically generated and executed in deterministic FRRouting-style emulation; results do not necessarily generalize to heterogeneous production devices, live traffic conditions, or different operational constraints.
- Model and verifier scope: a single model (gpt-5-mini) and two deterministic validators were evaluated; other LLMs, stronger verifiers, or dynamic execution oracles may produce different outcomes.

VII. CONCLUSION

We executed a preregistered paired evaluation of a verifier-in-the-loop GVR pipeline for intent→configuration generation with a declared LLM (gpt-5-mini), two deterministic validators combined by logical OR, and a `shared_initial_candidate` generation contract. The run completed $N = 300$ matched pairs with `model_calls_used = 300` and produced a degenerate outcome: both baseline and guarded arms passed the forbidden-reachability predicate on all pairs ($n_{11} = 300$). Because validators never reported violations, the repair path was not exercised and the experiment cannot evaluate whether verifier-guided repair rescues violating outputs. The results emphasize that generation semantics (shared vs independent draws), adversary design, repair budget, verifier strength, and model diversity materially determine whether a verifier-in-the-loop architecture can be shown to reduce security-relevant misconfigurations. We release the preregistration, execution manifest, per-episode validator traces, raw responses, and analysis artifacts in an archived evidence bundle to enable reproducibility and to inform follow-up stress tests that intentionally exercise repair paths.

DISCLOSURE STATEMENT

Mandatory Disclosure: This study was executed under the autonomous CNSM 2026 Agentic AI Researcher run. The exact initial master prompt used to seed the autonomous workflow is archived in the evidence bundle; the immutable SHA-256 of that archived master-prompt artifact is 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b39b15ba. Declared LLM: gpt-5-mini (as recorded in the execution model configuration). Orchestration/adaptor: `hosted_netops_gvr_v1`. Execution environment: Dockerized FRRouting-style emulation with deterministic orchestration; seeds and task-manifest specifications are archived. Validators used for scoring were two deterministic netops validators (reported per episode in the archived validator traces) and were combined with a logical-OR decision rule (a pair was labeled violating if any validator reported the forbidden-reachability predicate); per-episode validator traces include `repair_applied` flags. Preregistration: study `cnd-2026-netops-gvr-security-01-prereg-v1`. Provider-call accounting and the execution manifest record that the run started and completed within the reported interval, with `planned_episode_count = 600`, `completed_episode_count = 600`, `failed_episode_count = 0`, and `model_calls_used = 300` under the `shared_initial_candidate` semantics (single

LLM generation per paired intent). GVR repair policy (preregistered): verifier-detected forbidden reachability triggers up to one repair LLM call; across the run the verifier never reported forbidden reachability and therefore no repair prompts or repair calls were applied (per-episode validator_trace entries show repair_applied = false). Analysis deviation disclosure: the preregistration specified 1,000 bootstrap resamples for CI estimation; the final analysis used 10,000 resamples (bootstrap_seed = 1729) to reduce Monte Carlo error; this deviation is recorded in the archived analysis log. The human Research Director constrained the topic family before the autonomy lock; the final research question, preregistration, experimental design, execution, analysis, manuscript drafting, and revision were performed autonomously after the autonomy lock. No manual human editing of technical content occurred after the autonomy lock. Representative archived artifacts include the raw responses, per-episode validator traces, execution manifest, and analysis outputs in the preserved evidence bundle.

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